

Git Repository containing all the code: https://github.com/ktlevine/CS4420_Assignment05.git

How to compile the code:

- Gcc -Wall main.c -o main
- ./main <input_file> <algorithm> [quantum (only if needed)]

Test cases:

All outputs are based on the sample input provided in the instructions and on my git repository.

Running SJF:

```

kyle@kyle-QEMU-Virtual-Machine:~/OS/Assignment 5$ ./main input.txt SJF
The given processes are:
PID: 0 Arrival: 0 Burst: 12
PID: 1 Arrival: 2 Burst: 4
PID: 2 Arrival: 3 Burst: 1
PID: 3 Arrival: 4 Burst: 2

You have chosen the SJF Algorithm

Debugging info:
Process 0: Start: 0 End: 12 Waiting: 0 Running: 12
Process 2: Start: 12 End: 13 Waiting: 9 Running: 1
Process 3: Start: 13 End: 15 Waiting: 9 Running: 2
Process 1: Start: 15 End: 19 Waiting: 13 Running: 4

Final SJF Results:
PID  Arrival    Start    End    Running    Waiting
0      0         0      12      12         0
1      2        15      19       4        13
2      3        12      13       1         9
3      4        13      15       2         9
Average Waiting Time: 7.75

```

Running RR with a quantum of 5:

```

kyle@kyle-QEMU-Virtual-Machine:~/OS/Assignment 5$ ./main input.txt RR 5
The given processes are:
PID: 0 Arrival: 0 Burst: 12
PID: 1 Arrival: 2 Burst: 4
PID: 2 Arrival: 3 Burst: 1
PID: 3 Arrival: 4 Burst: 2

You have chosen the RR Algorithm

Debugging info:
Quantum: 5
Process 0: Start: 0 End: 5 Waiting: 0
Process 1: Start: 5 End: 9 Waiting: 3
Process 2: Start: 9 End: 10 Waiting: 6
Process 3: Start: 10 End: 12 Waiting: 6
Process 0: Start: 12 End: 17 Waiting: 0
Process 0: Start: 17 End: 19 Waiting: 7

Final RR Results:
PID  Arrival    Start    End    Running    Waiting
0      0         0      19      12         7
1      2         5       9       4         3
2      3         9      10       1         6
3      4        10      12       2         6
Average Waiting Time: 5.50

```

Running FCFS:

```
● kyle@kyle-QEMU-Virtual-Machine:~/OS/Assignment 5$ ./main input.txt FCFS
The given processes are:
PID: 0 Arrival: 0 Burst: 12
PID: 1 Arrival: 2 Burst: 4
PID: 2 Arrival: 3 Burst: 1
PID: 3 Arrival: 4 Burst: 2

You have chosen the FCFS Algorithm
=====
Debugging info:
Process 0: Start: 0 End: 12 Waiting: 0
Process 1: Start: 12 End: 16 Waiting: 10
Process 2: Start: 16 End: 17 Waiting: 13
Process 3: Start: 17 End: 19 Waiting: 13
=====
Final FCFS Results:
PID  Arrival    Start    End    Running    Waiting
0      0          0      12      12         0
1      2         12      16       4        10
2      3         16      17       1        13
3      4         17      19       2        13
Average Waiting Time: 9.00
```

Not giving a quantum value when trying to run RR:

```
● kyle@kyle-QEMU-Virtual-Machine:~/OS/Assignment 5$ ./main input.txt RR
The given processes are:
PID: 0 Arrival: 0 Burst: 12
PID: 1 Arrival: 2 Burst: 4
PID: 2 Arrival: 3 Burst: 1
PID: 3 Arrival: 4 Burst: 2

Invalid entry. Please enter in this format: <input_file> <algorithm> [quantum (only if needed)]
```

Trying to enter anything else besides what should be entered:

```
● kyle@kyle-QEMU-Virtual-Machine:~/OS/Assignment 5$ ./main input.txt testing random
The given processes are:
PID: 0 Arrival: 0 Burst: 12
PID: 1 Arrival: 2 Burst: 4
PID: 2 Arrival: 3 Burst: 1
PID: 3 Arrival: 4 Burst: 2

Invalid entry. Please enter in this format: <input_file> <algorithm> [quantum (only if needed)]
```